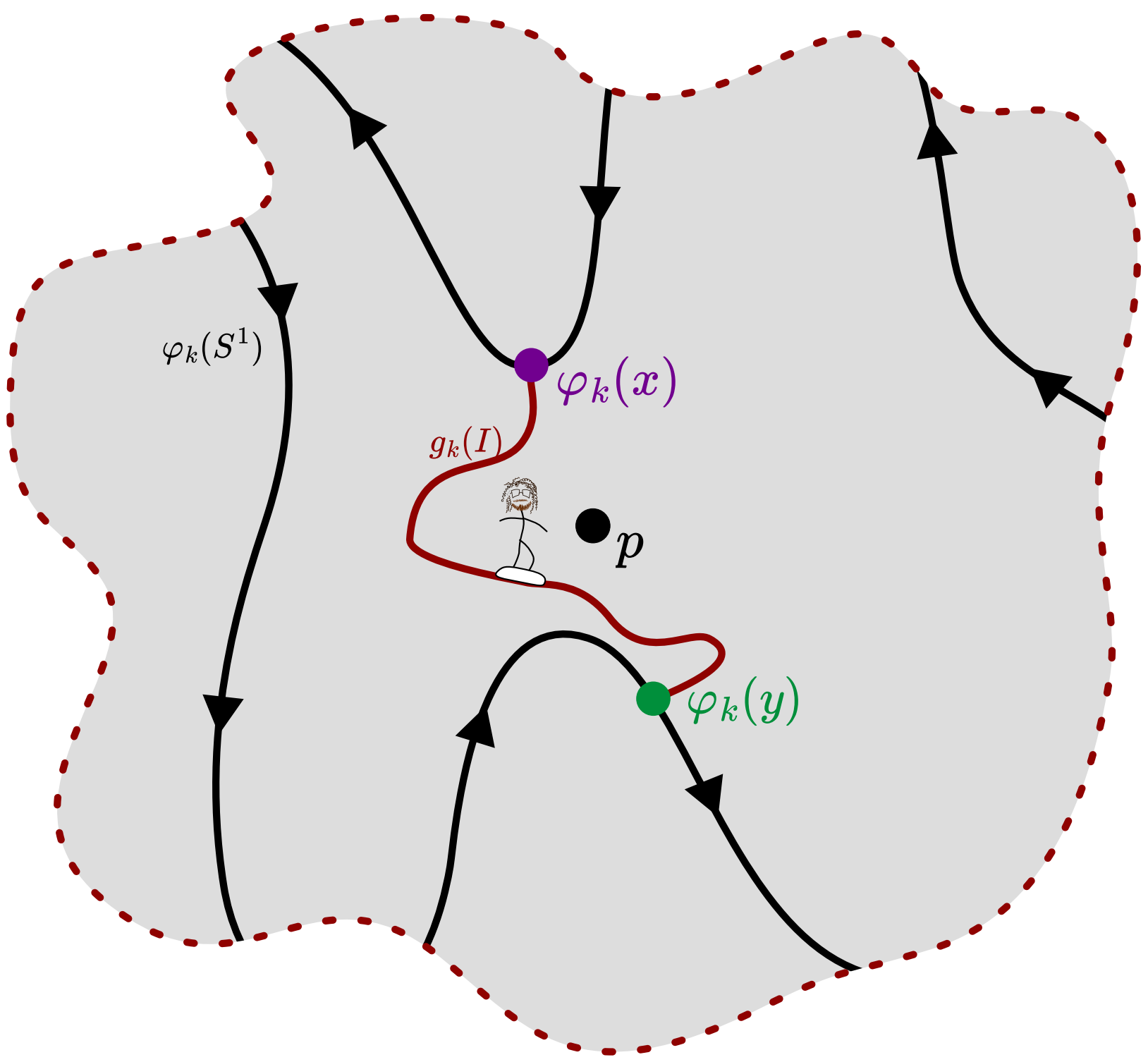

$\varphi_k(S^1)$
$\varphi_k(x)$
$g_k(I)$
p
$\varphi_k(y)$